MQTT & Data Specification Document

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Phase	Week 4
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1. Topic Hierarchy Architecture

StockSense utilizes a multi-tenant, hierarchical MQTT topic design. This pattern isolates data across organizational units while enabling granular broker-side access control (ACLs).

Canonical Pattern: `stocksense/{tenant_id}/{location_id}/{device_id}/{message_type}`

2. Topic Map & Routing Table

Topic Path Suffix	Dir	QoS	Purpose & Behavior Description
`.../telemetry`	NODE→CLOUD	QoS 1	Periodic status update (weights, metrics, battery, environmental values).
`.../events`	NODE→CLOUD	QoS 1/2	Asynchronous critical alerts (stock-out, unauthenticated removal, power failure).
`.../status`	NODE→CLOUD	QoS 1 (Retained)	Retained Will Topic payload maintaining edge connectivity state.
`.../cmd`	CLOUD→NODE	QoS 1	Inbound operational commands (tare scale, set interval, trigger reboot).
`.../cmd/response`	NODE→CLOUD	QoS 1	Execution acknowledgment and status output for issued commands.

3. Periodic Telemetry Payload (/telemetry)

Published continuously every 30 seconds (configurable) during normal operation.

```
{
  "timestamp": "2026-10-26T14:30:00Z",
  "device_id": "ESP32_NODE_0104",
  "firmware_version": "v1.2.4",
  "sequence_num": 14205,
  "metrics": {
    "raw_adc_counts": 8421050,
    "gross_weight_g": 2450.5,
    "net_weight_g": 1950.0,
    "unit_weight_g": 65.0,
    "calculated_qty": 30,
    "rfid_tag_id": "E2806894000050123456789A",
    "temperature_c": 24.6,
    "humidity_pct": 58.2
  },
  "power": {
    "mains_connected": true,
    "battery_voltage_v": 4.12,
    "battery_percentage": 98
  }
}
```

4. Asynchronous Event Payload (/events)

Published immediately upon state transitions, anomaly triggers, or threshold violations.

```
{
  "timestamp": "2026-10-26T14:32:15Z",
  "device_id": "ESP32_NODE_0104",
  "event_type": "ANOMALY_UNAUTHENTICATED_REMOVAL",
  "severity": "HIGH",
  "data": {
    "weight_before_g": 1950.0,
    "weight_after_g": 650.0,
    "delta_weight_g": -1300.0,
    "estimated_units_removed": 20,
    "rfid_present": false,
    "user_id": null,
    "action_recommended": "CHECK_CAMERA_OR_SHELF_LOCK"
  }
}
```

5. Event Enumerations

Event Code	Trigger Condition
STOCK_LOW	Calculated item count ≤ threshold minimum.
STOCK_OUT	Net weight reaches 0 g (± sensor noise floor).
ANOMALY_UNAUTHENTICATED_REMOVAL	Step weight drop without RFID/User authentication.
POWER_OUTAGE	External 5 V rail drop; switchover to battery supply.

6. Command & Control Payloads (/cmd & /cmd/response)

Issued by backend systems to execute scale operations or remote configuration.

```
{
  "command_id": "cmd_894321_tare",
  "action": "TARE_SCALE",
  "parameters": { "channel": "A", "known_reference_weight_g": 0.0 },
  "issued_by": "admin_kavindu",
  "timestamp": "2026-10-26T14:35:00Z"
}
```

7. Last Will & Testament Payload (/status)

```
{
  "device_id": "ESP32_NODE_0104",
  "status": "offline",
  "last_seen": "2026-10-26T14:40:12Z",
  "reason": "CONNECTION_LOST_OR_UNEXPECTED_SHUTDOWN"
}
```